Example 2.9 A girl A is reading a book having 446 pages and she has already finished reading 271 pages. She wants to finish reading this book within a week. What is the minimum number of pages she should read per day to complete reading the book within a week?

Solution:
Let x denote the number of pages the girl should read per day. Then we need our x to satisfy $7x + 271 \geq 446$. Hence $x \geq 25$; which implies that she should read at least 25 pages per day.

In all the above examples observe that each inequality has more than one solution. Inequalities in general give rise to a range of solutions.

Exercise - 2.3

1. Represent the following inequalities in the interval notation:
   (i) $x \geq -1$ and $x < 4$ (ii) $x \leq 5$ and $x \geq -3$
   (iii) $x < -1$ or $x < 3$ (iv) $-2x > 0$ or $3x - 4 < 11$.
2. Solve $23x < 100$ when (i) x is a natural number, (ii) x is an integer.
3. Solve $-2x \geq 9$ when (i) x is a real number, (ii) x is an integer, (iii) x is a natural number.
4. Solve: (i) $\frac{3(x-2)}{5} \leq \frac{5(2-x)}{3}$. (ii) $\frac{5-x}{3} < \frac{x}{2} - 4$.
5. To secure A grade one must obtain an average of 90 marks or more in 5 subjects each of maximum 100 marks. If one scored $84, 87, 95, 91$ in first four subjects, what is the minimum mark one scored in the fifth subject to get A grade in the course?
6. A manufacturer has 600 litres of a 12 percent solution of acid. How many litres of a 30 percent acid solution must be added to it so that the acid content in the resulting mixture will be more than 15 percent but less than 18 percent?
7. Find all pairs of consecutive odd natural numbers both of which are larger than 10 and their sum is less than 40.
8. A model rocket is launched from the ground. The height h reached by the rocket after t seconds from lift off is given by $h(t) = -5t^2 + 100t, 0 \leq t \leq 20$. At what time the rocket is 495 feet above the ground?
9. A plumber can be paid according to the following schemes: In the first scheme he will be paid rupees 500 plus rupees 70 per hour, and in the second scheme he will be paid rupees 120 per hour. If he works x hours, then for what value of x does the first scheme give better wages?
10. A and B are working on similar jobs but their monthly salaries differ by more than Rs 6000. If B earns rupees 27000 per month, then what are the possibilities of A's salary per month?

2.5 Quadratic Functions

In earlier classes we have learnt that for any $z \in \mathbb{R}$ and $n \in \mathbb{N}$, $z^n = z \cdot z \cdot z \cdots z$ (n-times).
A function of the form $P(x) = ax^2 + bx + c$, where $a, b, c \in \mathbb{R}$ are constants and $a \neq 0$, is called a quadratic function. If $P(t) = 0$ for some $t \in \mathbb{R}$, then we say t is a ***zero*** of $P(x)$.

